

SAFETY DATA SHEET

Print Date : 14.10.2022

1. PRODUCT AND COMPANY IDENTIFICATION

Product name : Engen Unleaded Petrol 95
CAS Identification : Mixture - Not Applicable

Supplier : Engen Petroleum Limited, Thibault Square, 8001, Cape Town (Tel: +27 (0) 21 403 4911, a/h (24/7): 0800 036 436

Emergency Response - South Africa : 086 100 0366 (Spill Tech 24/7); 0800 983 611 (CHEMTREC 24/7) and Poisons Information Helpline: 0861 555 777

Emergency Response - Africa : +44 20 3885 0382 (CHEMTREC 24/7)

Customer Service Centre : 0860 036 436 (Sales and Technical Information)

Engen Website : <http://www.engen.co.za/>

2. HAZARDS IDENTIFICATION

Emergency response data : Green Liquid. Product can accumulate a static charge and release vapours which may cause a fire or explosion. DOT ERG No. : 128

GHS Classification:

Health

Skin corrosion/irritation : Hazard Category 2.
Germ cell mutagenicity : Hazard category 1B.
Carcinogenicity : Hazard category 1A.
Reproductive toxicity : Hazard Category 2.
STOT - single exposure : Hazard category 3. Central nervous system depressant.
STOT - repeated exposure : Hazard Category 1
Aspiration hazard : Hazard Category 1

Environmental

Acute toxicity : Hazard category 2.
Chronic toxicity : Hazard category 2.

Physical

Flammability : Hazard category 1.

Signal Word : DANGER

GHS Labels/Pictograms:



Hazard Statements

Extremely flammable liquid and vapour. Causes skin irritation. May cause genetic defects. May cause cancer. Suspected of damaging fertility or the unborn child. May cause drowsiness or dizziness. May be fatal if swallowed and enters airways. Causes damage to organs through prolonged or repeated exposure. Toxic to aquatic life with long lasting effects.

Precautionary Statements

Engen Unleaded Petrol 95

Prevention

Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking. Keep container tightly closed. Ground/bond container and receiving equipment. Use explosion-proof electrical, ventilating and transfer equipment. Use non-sparking tools. Take action to prevent static discharges. Wear protective gloves, clothing, eye and face protection. Wash exposed skin after handling this product. Obtain, read and follow all safety instructions before use. Do not breathe dust/fume/gas/mist/vapours/spray. Use only outdoors or in a well-ventilated area. Avoid release to the environment.

Response

IN CASE OF FIRE: Use dry sand, dry chemical or alcohol-resistant foam to extinguish. IF ON SKIN (or hair): Immediately remove all contaminated clothing. Gently wash skin with plenty of soap and water. Launder contaminated clothing before re-use. If skin irritation occurs: Get medical attention. IF exposed or concerned: Get medical advice/ attention. IF INHALED: Remove to fresh air and keep at rest in a position comfortable for breath. Call a POISON CENTRE or doctor if you feel unwell. IF SWALLOWED: Immediately call a POISON CENTRE or doctor. Do NOT induce vomiting. Collect spillage.

Storage

Store in a well ventilated place. Keep container tightly closed. Keep cool. Store locked up.

Disposal

Dispose of contents/container to an approved waste disposal plant in accordance with local/regional/national/international regulation. Dispose of contents/container to an approved waste disposal plant in accordance with local/regional/national/international regulation.

See Section 11 for further health effects/toxicological data.

3. COMPOSITION/INFORMATION ON INGREDIENTS

Chemical name	CAS-No.	Weight%
Naphtha (petroleum) low boiling point	86290-81-5	> 80.00 - <= 95.00
Ethers, C5-6- branched alkyl Me	91995-60-7	>= 0.00 - < 5.00
Methyl tertiary butyl ether	1634-04-4	>= 0.00 - < 5.00
Benzene	71-43-2	<= 1.00
Toluene	108-88-3	>= 2.00 - <= 12.00
n-Hexane	110-54-3	>= 0.00 - <= 5.00

See Section 8 for Exposure Limits (if applicable).

4. FIRST AID MEASURES

Inhalation : Remove from further exposure. If respiratory irritation, dizziness, nausea, or unconsciousness occurs, seek immediate medical assistance. If breathing has stopped, assist ventilation with mechanical device or use mouth-to-mouth resuscitation with a mouthpiece.

Engen Unleaded Petrol 95

Skin contact	:	Remove contaminated clothing. Dry wipe exposed skin and cleanse with hand cleaner, soap and water. Launder contaminated clothing before reuse. (See Section 16 - Injection Injury)
Eye contact	:	Flush thoroughly with water. If irritation occurs call a doctor.
Ingestion	:	Seek immediate medical attention. Do not induce vomiting.
Note to doctors	:	Material if aspirated into the lungs may cause chemical pneumonitis. Skin contact may aggravate an existing dermatitis. Treat appropriately.

5. FIRE-FIGHTING MEASURES

Extinguishing media	:	Carbon dioxide, foam, dry chemical and water fog.
Special fire fighting procedure	:	Evacuate area. For large spills, fire fighting foam in sufficient quantities should be applied to blanket the flammable product surface.
Special protective equipment for firefighters	:	For fires in enclosed areas, fire fighters must use Self-Contained Breathing Apparatus.
Unusual fire and explosive hazards	:	EXTREMELY FLAMMABLE, HIGH HAZARD. Liquid can release considerable vapour at temperatures below ambient which readily form flammable mixtures. Vapours settle to ground level and may reach, via drains and other underground passages, ignition sources remote from the point of escape. Product can accumulate a static charge which may cause a fire or explosion.
Products of decomposition	:	Fumes, smoke, carbon monoxide, sulphur oxides, aldehydes and other decomposition products, in the case of incomplete combustion.
Flash Point	:	< -40 °C (ASTM D-56)
Upper Explosion Limit (UEL)	:	7.6 %(V)
Lower Explosion Limit (LEL)	:	1.4 %(V)
NFPA Hazard Id	:	Health: 1; Flammability: 3; Reactivity: 0

6. ACCIDENTAL RELEASE MEASURES

Procedure if material is released or spilled	:	Report spills/releases as required to appropriate authorities.
Methods for cleaning up	:	Eliminate sources of ignition. Warn occupants and/or ships in the downwind areas of fire and explosion hazard, and warn them to stay clear. LAND SPILL: Shut off source taking normal safety precautions. Take measures to minimize the effects on ground water. Recover by pumping using explosion-proof equipment or contain spilled liquid with sand or other suitable absorbent and remove mechanically into containers. If necessary, dispose of absorbed residues as directed in Section 13. WATER SPILL: Notify port and relevant authorities. Confine with booms if skimming equipment is available to recover the spill for later recycling or disposal.
Personal precautions	:	See Section 8.
Environmental precautions	:	Prevent spill from entering municipal sewers, water sources or low lying areas. Advise the relevant authorities if contaminations have occurred.

7. HANDLING AND STORAGE

Engen Unleaded Petrol 95

Safe handling advice	:	Use non-sparking tools and explosion-proof equipment. Never siphon by mouth. This product should not be used as a solvent or as a cleaning agent. Harmful in contact with or if absorbed through the skin. Avoid inhalation of vapours or mists. Use in well ventilated area away from all ignition sources. This liquid is volatile and gives off invisible vapours. Either the liquid or vapour may settle in low areas or travel some distance along the ground or surface to ignition sources where they may ignite or explode. Keep product away from high energy ignition sources, heat, sparks, pilot lights, static electricity, and open flames. It is unlawful and dangerous to put petrol into unapproved containers. Do not fill container in or on a vehicle. Static electricity may ignite vapours and cause fire. Place container on ground when filling and keep nozzle in contact with container. See Section 8 for additional personal protection advice when handling this product.
Storage information	:	This product is a static accumulator, therefore, all storage containers should be grounded and bonded. Drums should also be equipped with self-closing valves, pressure vacuum bungs and flame arresters. Outside or detached storage area, with an automatic sprinkling system, is preferred.
Storage and handling procedures	:	Electrical equipment and fittings must comply with local fire prevention regulations for this class of product. Refer to national or local regulations covering safety at petroleum handling and storage areas for this product.

8. EXPOSURE CONTROLS / PERSONAL PROTECTION

Occupational Exposure Limits (OELs)

Components	CAS-No.	Source	TWA	Value		Notations
Methyl tertiary butyl ether	1634-04-4	ACGIH TLV		180 mg/m ³	50 ppm	A3
Toluene	108-88-3	ACGIH TLV	LTEL		20 ppm	Skin; A4; BEI
Benzene	71-43-2	ACGIH TLV	LTEL STEL	1.6 mg/m ³ 8 mg/m ³	0.5 ppm 2.5 ppm	Skin; A1; BEI
n-Hexane	110-54-3	ACGIH TLV	LTEL	176 mg/m ³	50 ppm	Skin; BEI

LTEL: Long Term Exposure Limits - Time Weight Average (TWA) over 8 hours.

STEL: Short Term Exposure Limits - Time Weight Average (TWA) over 15 Minutes

Note: Limits Shown for guidance only. Follow applicable regulations.

Personal Protective Equipment (PPE)

Engineering controls	:	Use in well ventilated area. Explosive-proof ventilation equipment with local exhaust is desirable.
Respiratory protection	:	Approved respiratory equipment must be used when airborne concentrations are unknown or exceed the recommended exposure limit. Self-Contained Breathing Apparatus may be required for use in confined or enclosed spaces.

Engen Unleaded Petrol 95

Eye protection : If splash with liquid is possible, chemical type goggles should be worn.

9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance : Liquid.
Colour : Green
Odour : Hydrocarbon
Boiling point : > 30 - 210 °C
Flash Point : < -40 °C (ASTM D-56)
Flammability : NA
Upper Explosion Limit (UEL) : 7.6 %(V)
Lower Explosion Limit (LEL) : 1.4 %(V)
Vapour pressure : 542 hPa
Relative vapour density : 3
Density : >= 0.710 g/cm³ @ 20 °C (ASTM D-4052) <= 0.7850 g/cm³ @ 20 °C (ASTM D-4052)

Viscosity, kinematic : < 1 mm²/s @ 40 °C (ASTM D-445)
< 0.1 mm²/s @ 100 °C

10. STABILITY AND REACTIVITY

Stability : Stable.

Conditions to avoid : Heat, sparks, flame and build up of static electricity.

Materials to avoid : Halogens, strong acids, alkalis and oxidizers.

Hazardous decomposition products : Fumes, smoke, carbon monoxide, sulphur oxides, aldehydes and other decomposition products, in the case of incomplete combustion.

11. TOXICOLOGICAL INFORMATION

Acute toxicity

Components

Naphtha (petroleum) low boiling point

Acute oral toxicity : No available data.
Acute inhalation toxicity : No available data.
Acute dermal toxicity : No available data.

Ethers, C5-6- branched alkyl Me

Acute oral toxicity : Harmful if swallowed.
Acute inhalation toxicity : No available data.
Acute dermal toxicity : No available data.

Methyl tertiary butyl ether

Acute oral toxicity : No available data.
Acute inhalation toxicity : No available data.
Acute dermal toxicity : No available data.

Benzene

Acute oral toxicity : LD50 (Rats) : > 2,000 mg/kg
Acute inhalation toxicity : LC50 (Rats) : 43.7 mg/l
Acute dermal toxicity : LD50 (Rabbits) : > 8,260 mg/kg

Toluene

Engen Unleaded Petrol 95

Acute oral toxicity	:	LD50 (Rats) : > 5,000 mg/kg
Acute inhalation toxicity	:	(Rats) : > 20 mg/l
Acute dermal toxicity	:	(Rabbits) : 12,267 mg/kg

n-Hexane

Acute oral toxicity	:	No available data.
Acute inhalation toxicity	:	No available data.
Acute dermal toxicity	:	No available data.

Skin corrosion/irritation Components

Naphtha (petroleum) low boiling point

Result	:	Causes skin irritation.
--------	---	-------------------------

Ethers, C5-6- branched alkyl Me

Result	:	No available data.
--------	---	--------------------

Methyl tertiary butyl ether

Result	:	Causes skin irritation.
--------	---	-------------------------

Benzene

Species	:	Rabbits
Exposure time	:	4 h
Result	:	Causes skin irritation.

Toluene

Species	:	Rabbits
Exposure time	:	4 h
Result	:	Causes skin irritation.

n-Hexane

Result	:	Causes skin irritation.
--------	---	-------------------------

Eye irritation Components

Naphtha (petroleum) low boiling point

Result	:	No known significant effects or critical hazards.
--------	---	---

Ethers, C5-6- branched alkyl Me

Result	:	No known significant effects or critical hazards.
--------	---	---

Methyl tertiary butyl ether

Result	:	No available data.
--------	---	--------------------

Benzene

Species	:	Rabbits
Result	:	Causes serious eye irritation.

Toluene

Species	:	Rabbits
Result	:	(Rabbits): Slightly irritating.

n-Hexane

Result	:	No known significant effects or critical hazards.
--------	---	---

Sensitization

Engen Unleaded Petrol 95

Components

Naphtha (petroleum) low boiling point

Remarks : No known significant effects or critical hazards.

Ethers, C5-6- branched alkyl Me

Remarks : No known significant effects or critical hazards.

Methyl tertiary butyl ether

Remarks : No available data.

Benzene

Remarks : No available data.

Toluene

Remarks : No available data.

n-Hexane

Remarks : This product is not a skin sensitizer.

Germ cell mutagenicity Components

Naphtha (petroleum) low boiling point

Remarks : May cause genetic defects.

Ethers, C5-6- branched alkyl Me

Remarks : No known significant effects or critical hazards.

Methyl tertiary butyl ether

Remarks :

Benzene

Remarks : May contain benzene /constituent classified as germ cell mutagens

Toluene

Remarks : No available data.

n-Hexane

Remarks : No known significant effects or critical hazards.

Carcinogenicity Components

Naphtha (petroleum) low boiling point

Remarks : May cause cancer.

Ethers, C5-6- branched alkyl Me

Remarks : Not classifiable as a human carcinogen.

Methyl tertiary butyl ether

Engen Unleaded Petrol 95

Remarks : No available data.

Benzene

Remarks : Known human carcinogen - may cause leukemia

Toluene

Remarks : No available data.

n-Hexane

Remarks : Not classifiable as a human carcinogen.

Reproductive toxicity Components

Naphtha (petroleum) low boiling point

Remarks : Suspected human reproductive toxicant.

Ethers, C5-6- branched alkyl Me

Remarks : No known significant effects or critical hazards.

Methyl tertiary butyl ether

Remarks : No available data.

Benzene

Remarks : No available data.

Toluene

Remarks : Suspected human reproductive toxicant.

n-Hexane

Remarks : Suspected human reproductive toxicant.

STOT - single exposure Components

Naphtha (petroleum) low boiling point

Remarks : May cause drowsiness or dizziness.

Ethers, C5-6- branched alkyl Me

Remarks : May cause drowsiness or dizziness.

Methyl tertiary butyl ether

Remarks : No available data.

Benzene

Remarks : No available data.

Toluene

Remarks : May cause drowsiness or dizziness. Central nervous system depressant.

n-Hexane

Remarks : May cause drowsiness or dizziness.

Specific target organ toxicity (STOT) - repeated exposure Components

Naphtha (petroleum) low boiling point

Engen Unleaded Petrol 95

Remarks : No known significant effects or critical hazards.

Ethers, C5-6- branched alkyl Me

Remarks : No known significant effects or critical hazards.

Methyl tertiary butyl ether

Remarks : No available data.

Benzene

Target Organs

Remarks : (Thymus, Liver, Bone Marrow)
: Causes damage to organs through prolonged or repeated exposure. Central nervous system.

Toluene

Remarks : May cause damage to organs. Central nervous system.

n-Hexane

Remarks : May cause damage to organs (central nervous system/ peripheral nervous system) Central nervous system.

Aspiration hazard Components

Naphtha (petroleum) low boiling point

Remarks : May be fatal if swallowed and enters airways.

Ethers, C5-6- branched alkyl Me

Remarks : No available data.

Methyl tertiary butyl ether

Remarks : No available data.

Benzene

Remarks : May be fatal if swallowed and enters airways.

Toluene

Remarks : May be fatal if swallowed and enters airways.

n-Hexane

Remarks : May be fatal if swallowed and enters airways.

12. ECOLOGICAL INFORMATION

Ecotoxicity effects Components

Naphtha (petroleum) low boiling point

Acute toxicity : No available data.

Chronic toxicity : Fish, invertebrates and algae with LC/EC50 >1 - 10 mg/L

Ethers, C5-6- branched alkyl Me

Acute toxicity : No available data.

Chronic toxicity : No available data.

Methyl tertiary butyl ether

Engen Unleaded Petrol 95

Acute toxicity : No available data.
Chronic toxicity : No available data.

Benzene

Acute toxicity : Fish (Oncorhynchus mykiss) 96h LC50 5.3 mg/L
Invertebrates (Daphnia magna) 48h EC50 10 mg/L
Algae (Pseudokirchnerella subcapitata) 72h ErC50 100 mg/L
Chronic toxicity : Fish (Pimephales promelas) 32d NOEC 0.8 mg/L
Invertebrates (Ceriodaphnia dubia) 7d NOEC 3 mg/L

Toluene

Acute toxicity : Fish (Oncorhynchus kisutch) 96h LC50 5.5 mg/L
Invertebrates (Ceriodaphnia dubia) 48h LC50 3.78 mg/L
Algae (Chlamydomonas angulosa) 3h EC50 (photosynthesis rate) 134 mg/L
Chronic toxicity : Fish (Oncorhynchus kisutch) 40d NOEC 1.39 mg/L

n-Hexane

Acute toxicity : No available data.
Chronic toxicity : Fish, invertebrates and algae with LC/EC50 >1 - 10 mg/L

Persistence and degradability Components

Naphtha (petroleum) low boiling point

Biodegradability : Not established.

Ethers, C5-6- branched alkyl Me

Biodegradability : This product is expected to be inherently biodegradable at a slow to moderate rate.

Methyl tertiary butyl ether

Biodegradability : This product is not readily biodegradable.

Benzene

Biodegradability : Readily Biodegradable.

Toluene

Biodegradability : Readily Biodegradable.

n-Hexane

Biodegradability : This product is expected to be inherently biodegradable.

Bioaccumulation Components

Naphtha (petroleum) low boiling point

Bioaccumulation : Not established.

Ethers, C5-6- branched alkyl Me

Bioaccumulation : Not established.

Methyl tertiary butyl ether

Engen Unleaded Petrol 95

Bioaccumulation : Not established.

Benzene

Bioaccumulation : Does not bio-accumulate significantly.

Toluene

Bioaccumulation : BCF = 90

n-Hexane

Bioaccumulation : Does not bio-accumulate significantly.

13. DISPOSAL CONSIDERATIONS

Waste disposal : Product is suitable for burning for fuel value in compliance with applicable laws and regulations, and consideration of product characteristics at time of disposal.

Contaminated packaging : Empty containers retain residue (liquid and/or vapour) and can be dangerous. DO NOT PRESSURIZE, CUT, WELD, BRAZE, SOLDER, DRILL, GRIND OR EXPOSE SUCH CONTAINERS TO HEAT, FLAME, SPARKS, STATIC ELECTRICITY, OR OTHER SOURCES OF IGNITION; THEY MAY EXPLODE AND CAUSE INJURY OR DEATH. Do not attempt to refill or clean container since residue is difficult to remove.

Other regulations : This material and any waste emanating from its use must be disposed via an appropriate method in compliance with the applicable National, regional and local legislative/ authority requirements.

Flash Point : < -40 °C (ASTM D-56)

14. TRANSPORT INFORMATION

ADR

Proper shipping name : GASOLINE
UN number : 1203
Class : 3
Letter : F
Packing group : I
Labelling number : 3

CFR

Proper shipping name : GASOLINE
UN number : 1203
Class : 3
Letter : F
Packing group : I
Labelling number : 3

IATA_C

Proper shipping name : GASOLINE
UN number : 1203
Class : 3
Letter : F
Packing group : I
Labelling number : 3

IMDG

Engen Unleaded Petrol 95

Proper shipping name : GASOLINE
UN number : 1203
Class : 3
Letter : F
Packing group : I
Labelling number : 3
Marine pollutant : Marine Pollutant
Medical First Aid Guide (MFAG) table : 311
Emergency Schedule (EmS) number : 3-07
IMDG code page number : 3141

15. REGULATORY INFORMATION

South African Legislation and Standards : South African OHS Act, 85 of 1993: Hazardous Chemical Agents, Regulation 14A

National Environmental Management: Waste Act 2008

South African Guidelines

SANS 10234:2008 - Globally Harmonized System of classification & labelling of chemicals

SANS 11014:2010 - Safety data sheet for chemical products - Content & order of sections South African OHS Act, 85 of 1993: Hazardous Chemical Agents, Regulation 14A

National Environmental Management: Waste Act 2008

South African Guidelines

SANS 10234:2008 - Globally Harmonized System of classification & labelling of chemicals

SANS 11014:2010 - Safety data sheet for chemical products - Content & order of sections

Other regulations : Globally Harmonized System of Classification and Labelling of Chemicals (GHS)

The components of this product are reported in the following inventories:

.

16. OTHER INFORMATION

INJECTION INJURY WARNING: If product is injected into or under the skin, or into any part of the body, regardless of the appearance of the wound or its size, the individual should be evaluated immediately by a doctor as a surgical emergency. Even though initial symptoms from high pressure injection may be minimal or absent, early surgical treatment within the first few hours may significantly reduce the ultimate extent of injury.

Abbreviations That May Have Been Used In This Document:

TLV - Threshold Limit Value	TWA - Time Weighted Average
LTTEL - Long-term Exposure Limit	STEL - Short-term Exposure Limit
ACGIH - American Conference of Governmental Industrial Hygienists	CAS - Chemical Abstract Service Number

Engen Unleaded Petrol 95

IMO/IMDG - International Maritime Dangerous Goods Code	SDS – Safety Data Sheet
SANS – South African National Standards	GHS - Globally Harmonized System of Classification and Labelling of Chemicals
STOT – Specific Target Organ Toxicity	OHS Act – Occupational Health and Safety Act
NFPA - National Fire Protection Association (USA)	ECHA - European Chemicals Agency

Sources of key data used to compile the Safety Data Sheet : European Oil Company Organisation for Environment, Health and Safety)
ECHA (<https://echa.europa.eu/information-on-chemicals>)

Original Release : 04.10.2021
Last Date of Review : 14.10.2022

Disclaimer

Information given herein is offered in good faith as accurate, but without guarantee. Conditions of use and suitability of the product for particular uses are beyond our control; all risks of use of the product are therefore assumed by the user and we expressly disclaim all warranties of every kind and nature, including warranties of merchantability and fitness for a particular purpose in respect to the use or suitability of the product. Nothing is intended as a recommendation for uses which infringe valid patents or as extending license under valid patents. Appropriate warnings and safe handling procedures should be provided to handlers and users.
